Filbin 2018 pediatric H3-K27M glioma — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

All 6 SmartSeq2 samples intermix on one connected kNN component (no per-patient archipelago); a coherent cycling lobe (261 cells) is scoped out so the fork resolves.

Gate 2 — module geography → GO (branching)

Gast/pSom co-express at the Mixed hub and diverge into two arms; Crest peaks at the OC-arm tip (terminal readout). Branching module pattern present.

Gate 3 — cross-repression → GO

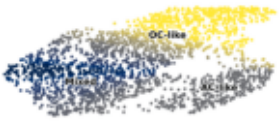
Mixed hub (n=761) per-gene Gast↔pSom r = −0.34 (anchor Puram −0.49); permutation null +0.00 ± 0.09, p<0.001 → real competitive cross-repression, not a collider artifact.

STEP FIGURES

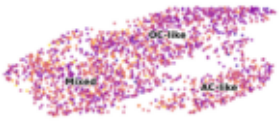
Manifold — samples



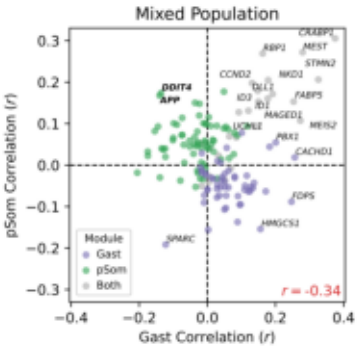
Trajectory / root



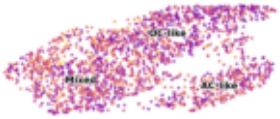
Gastrulation



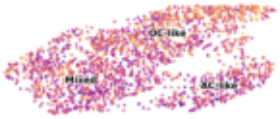
Gate 3: r = −0.34



Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Pediatric H3-K27M diffuse midline glioma (Filbin et al., 2018), SmartSeq2 (TPM) — 2,259 malignant cells, 6 samples; 1,998 in the developmental fork. Branching (hub-and-spoke).

Verdict.
Branching — the embryonic pattern reproduces: a Gast/pSom Mixed hub diverges into AC-like and OC-like arms with cross-repression (r=−0.34); Crest is a terminal readout at the OC tip. All three gates GO. EMT-TFs PRRX1 / ZEB1 (with ZEB2).

Parameter choices

Min-cell threshold	≥100 (no clean count gap; keeps all 6)	**Integration**	Harmony on sample, θ=10
Embedding	PCA50 → n_pcs=20, k=30, UMAP spread=1.3	**Overdispersed genes**	3,779
Cycling lobe removed	Leiden clusters (>50% cyc), 261 cells	**Trajectory**	scFates PPT tree (Nodes=40, λ=3000)
Root	Gast/pSom-high, Crest-low tip (non-Crest, not the fork)	**Hub / EMT-TFs**	Mixed convergence cluster (n=761); PRRX1 / ZEB1 / ZEB2